

Elimination Reactions of Halogenoalkanes (Cambridge (CIE) A Level Chemistry): Revision Note

Elimination Reactions of Halogenoalkanes

- An elimination reaction involves the loss of a small molecule from a larger organic molecule
- In **halogenoalkanes**, this small molecule is usually a **hydrogen halide** (e.g. HBr or HCl)
- The product is typically an **alkene**
 - The elimination reaction of bromoethane with ethanolic sodium hydroxide

Reaction Conditions for Elimination

- Halogenoalkanes are heated with **ethanolic sodium hydroxide** (NaOH dissolved in ethanol)
- Under these **anhydrous conditions**, elimination occurs:
 - The **C-X bond** (where X = halogen) breaks **heterolytically**
 - A **halide ion (X⁻)** is released
 - A **double bond** forms, producing an **alkene**
- E.g. Elimination of bromoethane:

bromoethane + sodium hydroxide (ethanol) → ethene + sodium bromide + water



- One **hydrogen atom** and the **bromine atom** are eliminated
- The carbon chain forms a **C=C double bond** to make **ethene**



Examiner Tips and Tricks

Reaction **conditions are crucial** in determining the product.

Elimination = hot, ethanolic NaOH

Substitution = warm, aqueous NaOH